Lab Exercise 2: Filipino Hate Speech Classification Analysis

1. Exploratory Data Analysis

1.1 Data Distribution - To understand the dataset balance, I visualized the count of "Hate" (1) versus "Non-Hate" (0) samples.

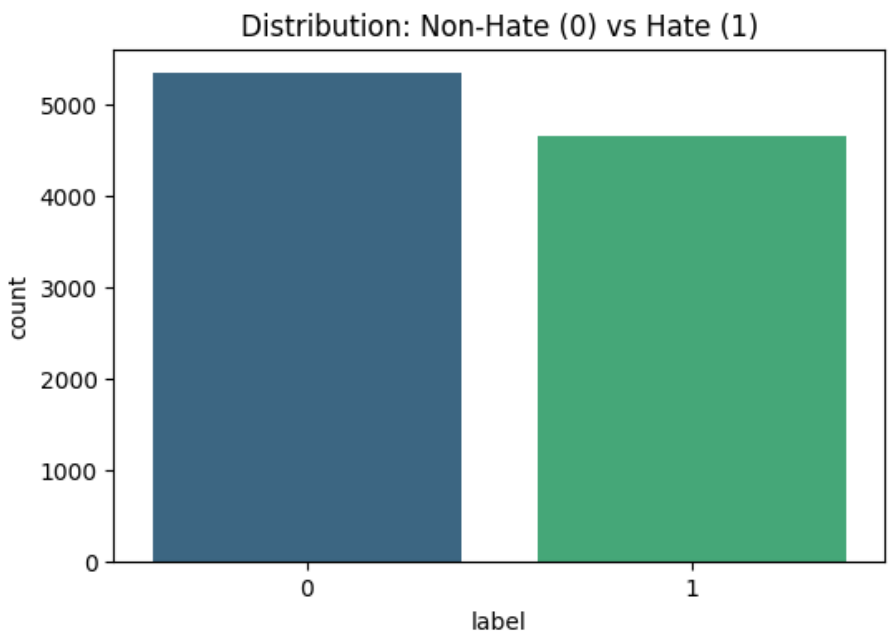


Figure 1: Distribution of classes in the training set.

The training dataset is relatively balanced, though there is a slight skew towards the "Non-Hate" class. As seen in Figure 1, there are approximately 5,300 Non-Hate samples (Label 0) compared to approximately 4,700 Hate samples (Label 1).

This slight imbalance is not severe enough to require major resampling techniques, but it implies that the baseline accuracy (guessing the majority class) would be slightly above 50%.

1.2 Feature Analysis (Top Predictive Words) - I analyzed the predictive strength of words using the model's coefficients. This reveals which words actually drive the model to predict "Hate" vs "Non-Hate."

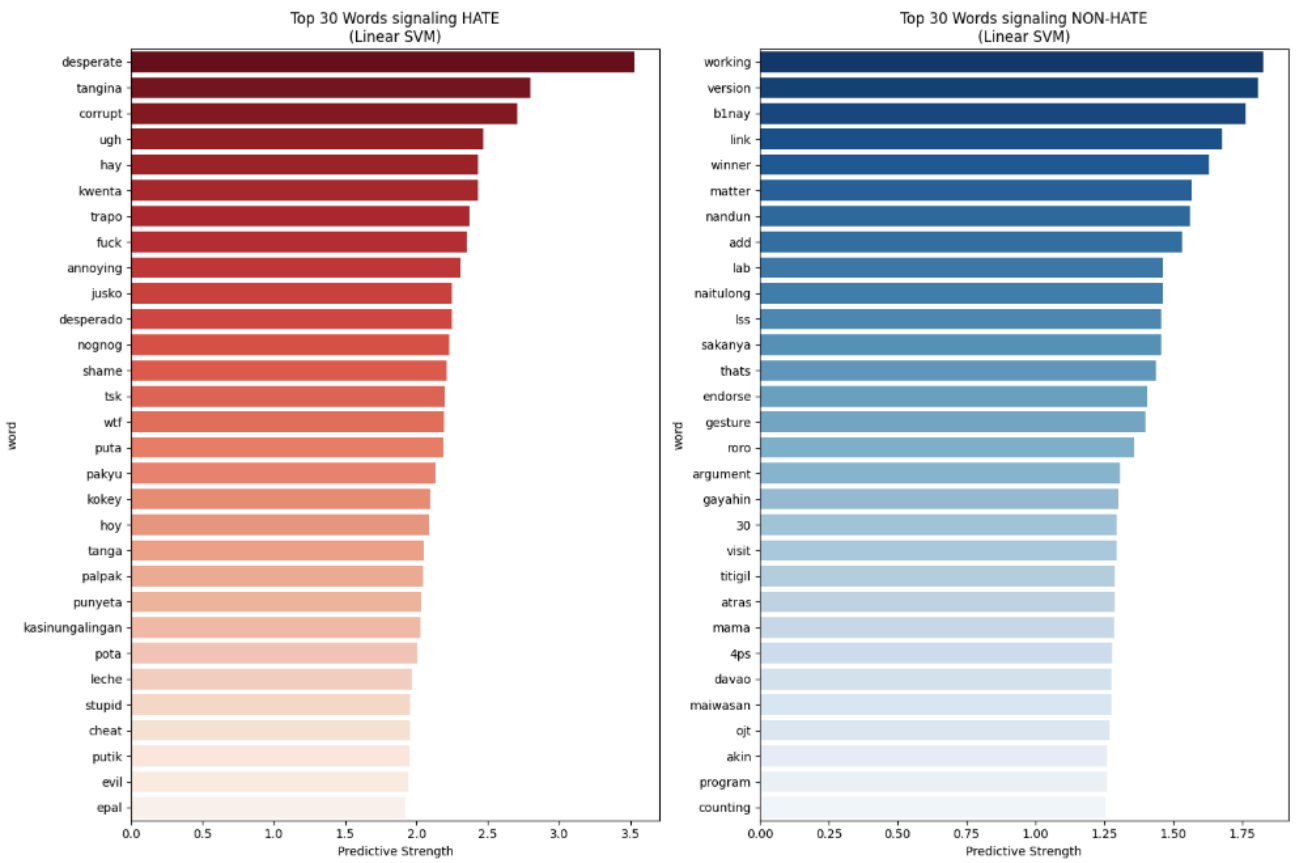


Figure 2: Top 30 words with highest predictive strength for each class.

Hate Speech Indicators:

- **Key Words:** "desperate", "tangina", "corrupt", "trapo", "fuck", "nognog", "puta".
- The strongest indicators for hate speech are explicitly negative. They fall into three categories:
    1. **Profanity:** "tangina", "fuck", "puta".
    2. **Political Insults:** "trapo" (traditional politician), "corrupt", "nognog" (a colorist slur).
    3. **Emotional Outbursts:** "ugh", "hay", "annoying".
- This suggests the model relies heavily on **lexical matching o**f profanity to detect hate.

Non-Hate Indicators:

- **Key Words:** "working", "version", "link", "winner", "matter", "naitulong".
- **Observations:** The non-hate vocabulary is much more functional. Words like "link", "add", and "version" suggest technical discussions. Interestingly, the word "b1nay" appears here, suggesting that this specific spelling variation might be used in less toxic contexts compared to slurs like "nognog."

2. Model Performance Summary

I trained three models using TF-IDF vectorization. The random state was set to my student number for reproducibility.

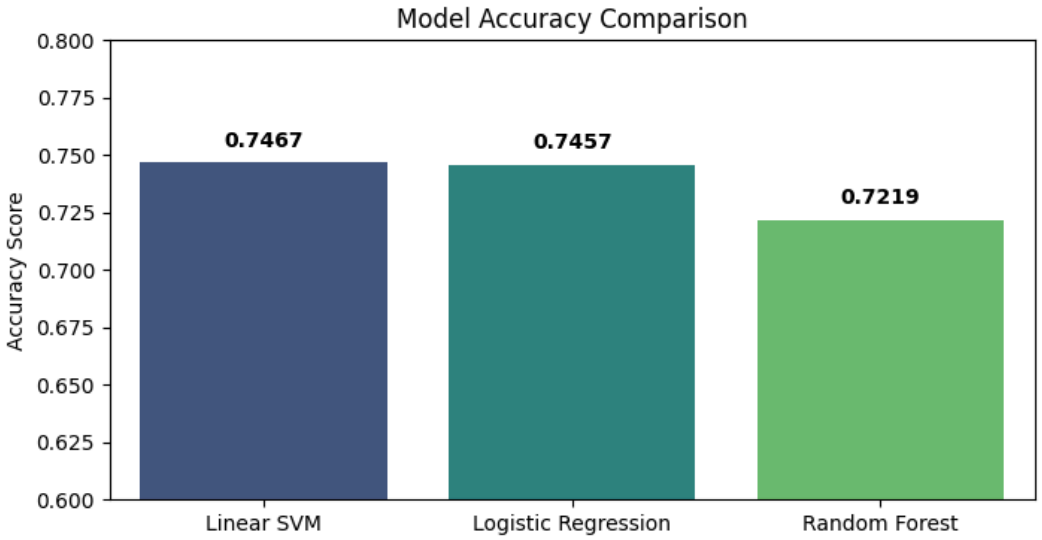


Figure 3: Accuracy comparison across three models.

Model	***Class***	***Precision***	***Recall***	***F1-Score***
Linear SVM	*Non-Hate (0)*	*0.74*	*0.79*	*0.77*
Accuracy: 0.7467	*Hate (1)*	*0.75*	*0.70*	*0.72*
Logistic Regression	*Non-Hate (0)*	*0.74*	*0.79*	*0.77*

Accuracy: 0.7457	*Hate (1)*	*0.75*	*0.70*	*0.72*
Random Forest	*Non-Hate (0)*	*0.72*	*0.76*	*0.74*
Accuracy: 0.7219	*Hate (1)*	*0.72*	*0.68*	*0.70*

Table 1: Detailed Classification Report for the trained models.

Summary of Results:

Model Algorithm	**Accuracy Score**	**Rank**
Linear SVM	0.7467	Highest
Logistic Regression	0.7457	Middle
Random Forest	0.7219	Lowest

Table 2: Detailed Summary Results of the Models

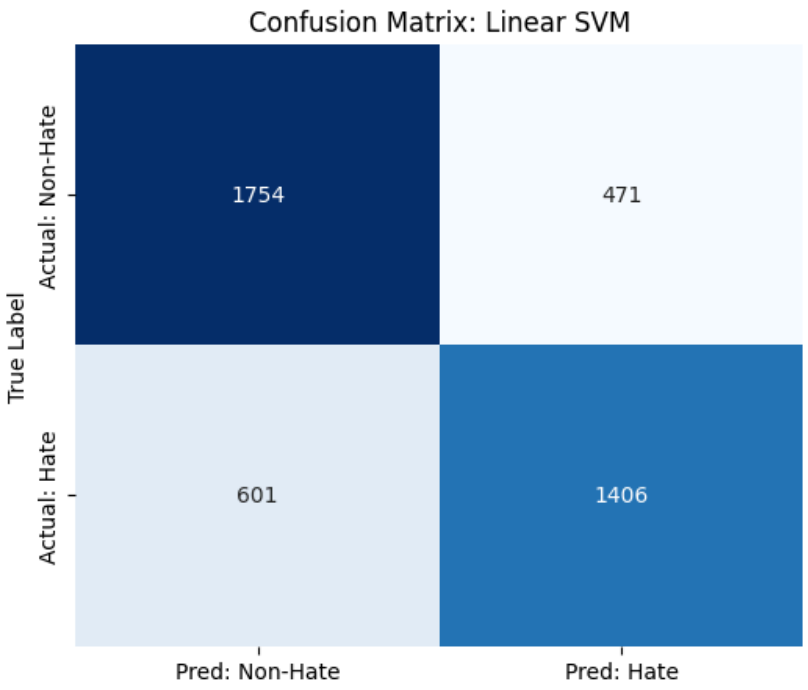
- **Highest Performing Model: Linear SVM (74.67%)**
- **Lowest Performing Model: Random Forest (72.19%)**

The Linear SVM and Logistic Regression performed almost identically. This is common for high-dimensional sparse text data (TF-IDF), where linear boundaries often work better than complex decision trees. Random Forest likely struggled due to the sparsity of the data.

3. Analysis of Hard-to-Classify Samples

This section analyzes samples that were misclassified by the best performing model (Linear SVM)

3.1 Confusion Matrix Analysis



Looking at the confusion matrix above:

- **True Negatives (Correct Non-Hate):** 1754
- **True Positives (Correct Hate):** 1406
- **False Positives (Safe text flagged as Hate):** 471
- **False Negatives (Hate Speech missed by model):** 601

The model has a higher number of False Negatives (601) than False Positives. This means the model is more likely to miss hate speech(predicting it as safe than it is to falsely accuse someone of hate speech.

3.2 Qualitative Analysis of Misclassified Texts

I examined the "Hard-to-Classify" samples where the model failed.

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PART 4: HARD TO CLASSIFY SAMPLES
Number of 'Hard' Samples: 679

--- Example Hard Cases ---
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
text  \
0         Unshaded votes and votes for Mayor Duterte goes to Mar Roxas according to some reports of ballot tests . [HASHTAG]
1                                                                                                                Na-Binay ????? ? [HASHTAG] [LINK]
6                                                                Binay in Isabela radio interview : ‘Yong iba , nangangako pa . Nagawa ko na ho lahat ‘yon . [HASHTAG]
18  [MENTION] [MENTION] I will support your decisions babies , but sorry to say , Mar Roxas is not fit to be president ! ? ?
21                                                                                                                hay nako basta ako BINAY ako . BINAY tayo . hahaha

    label  Pred_Best  Pred_Worst
0       1          0           0
1       1          0           0
6       0          1           1
18      1          0           0
21      0          1           1
```

Type A: False Negatives (Hate Speech predicted as Non-Hate)

Sample: *"Unshaded votes and votes for Mayor Duterte goes to Mar Roxas according to some reports..."*

This is disinformation/conspiracy. It contains no profanity or direct insults ("bobo"). The model sees neutral words ("votes", "reports") and classifies it as safe, missing the malicious intent.

Sample: *"[MENTION] ... I will support your decisions babies , but sorry to say , Mar Roxas is not fit to be president ! ? ?"*

The sentence structure is polite ("support your decisions", "sorry to say"). The hate sentiment is buried under positive phrasing. The TF-IDF model likely weighted "support" as a positive feature, canceling out the negative sentiment.

Type B: False Positives (Safe text predicted as Hate)

Sample: *"Binay in Isabela radio interview : 'Yong iba , nangangako pa . Nagawa ko na ho lahat 'yon..."*

This is a quote from a politician. However, the model flagged it as Hate likely because it contains the word "Binay". In the training set, "Binay" is likely heavily correlated with hate speech labels. The model is guilty of associative bias, assuming any mention of the target entity is hateful.

3.3 Conclusion on "Hard" Samples

The hard-to-classify samples reveal a limitation in the TF-IDF + SVM approach. The model relies heavily on keywords (profanity = hate). It fails when:

- Hate is expressed without profanity (conspiracy theories).
- Hate uses context-dependent slang or sarcasm.
- Neutral texts contain "trigger words" (names of controversial politicians).

4. Overall Conclusion

In this laboratory exercise, I developed and evaluated three machine learning models to classify Filipino hate speech. The results demonstrated that a Linear Support Vector Machine (SVM) utilizing TF-IDF vectorization was the most effective approach, achieving a testing accuracy of 74.67%. This performance slightly outperformed Logistic Regression and notably surpassed the Random Forest classifier.

The Exploratory Data Analysis (EDA) and subsequent error analysis revealed that the models are highly effective at detecting explicit hate speech characterized by profanity and direct insults (e.g., "tangina," "bobo"). However, the reliance on TF-IDF (a frequency-based feature) created a significant limitation: the models lack contextual understanding. This resulted in distinct failure modes where "clean" hate speech (such as sarcasm, disinformation, or conspiracy theories without swear words) was missed, and neutral statements containing controversial entities (like "Binay") were incorrectly flagged as hate due to associative bias.

To improve performance in future iterations, relying solely on keyword statistics is insufficient for the nuances of the Filipino language. Future work should transition from frequency-based techniques like TF-IDF to semantic vector representations (such as Word2Vec or BERT). Unlike TF-IDF, which treats words as independent units, these advanced models can capture the sequence, context, and sentiment of a sentence, allowing for better detection of sarcasm and indirect hate speech.